



(Image Source: www.cptgroup.cn)

several traffic rules and put fellow travellers' lives at risk. None of this is acceptable. However, this may happen if your AI is too narrow and cannot see the forest of trees.

Many data with only a few types of scenarios cannot train the solution for real-life use cases, and this is a significant risk. Unless your training dataset represents all the possible use cases and scenarios, it is useless.

Since ML is an inductive process, the base model used by your solution can only cover what it has been trained on and seen in the data. If your training has missed low occurrence scenarios (also known as long-tail or edge cases), your solution will not be able to support them. Your solution will be too risky in those scenarios or may fail.

If the data is missing variety, it can lead to several problems in the future. The solution developed with a limited dataset and narrow training would be highly risky. Also, if the solution design has covered the breadth and depth of data and as many scenarios as reasonably possible, risks are lower.

4. Is there an option to stop gracefully?

A system that is designed right will always have an inbuilt redundancy. New-age AI solutions can be highly potent and may result in scalable damage when something goes wrong.

So, ask your development team, or your AI vendors, about the kill-switch. Check with them the possibility of stopping the system gracefully when needed.

A system without an option to stop gracefully is a highly risky system.

You may not have an option to stop the system when needed. However, exercising this option is not as simple as pulling the plug. This means you may still face a significant risk of jeopardising business continuity in the process if you decide to stop it.

The ability to stop is not as bad as not being able to stop at all, though. In this case, you must evaluate your options and alternate plan. What are the financial and overall business repercussions of those alternative options when invoked? If stopping the AI system in an emergency would mean you are required to put up a hundred-member team in place to cope with the situation, it could be a significant financial and HR burden for you.

When evaluating your AI solution on this parameter, check the availability of kill switch or graceful-stop and restart options. If there are no such options, it is a high-risk solution. However, if these options are available with some level of inconvenience and minor losses, it may be termed as a medium-risk solution. Although highly unlikely, there may be a solution that is easy to stop and restart. If it will not cost you too many resources, it may be a low-risk option and should be the most preferred one from a risk point of view.

If you cannot stop an AI solution easily and gracefully, it is a high-risk solution. On the contrary, if you are in full control and can exercise that option easily and without any significant loss, it is low risk, and is a better solution.

5. Is the solution explainable and auditable?

A software solution, no matter how sophisticated, is always auditable,

understandable, and explainable.

Specific to the AI solution, explainability has been one of the most talked-about topics. With so many different approaches to develop AI solutions, it is getting increasingly difficult to determine how systems are making decisions. With explainable and auditable AI solutions, we should be able to fix accountability, make corrections, and take corrective actions as well.

For many industries, AI explainability is a regulatory requirement. For example, with the General Data Protection Regulation (GDPR) laws in effect, companies will be required to provide consumers with an explanation for AI-based decisions.

There are growing risks of AI when implemented without adequate explainability. These risks affect ethics, business performance, regulation, and our ability to learn iteratively from AI implementations.

People do not want to accept arbitrary decisions made by some entity that they don't understand. To be acceptable and trustworthy, you would want to know why the system is making a specific decision.

In addition to explainability, being auditable is another key to the right AI solution. Whenever something goes wrong, you would need to recover an audit trail of the entire transaction. Without being auditable, AI solutions are a significant risk. In the absence of accurate records, regulatory fines may ramp up, and financial losses could pile up quickly.

Not only this, but it also becomes a limiting factor for you as a business. Without proper explanation and understanding, you cannot improve. Controlled and measurable improvements become near impossible when you are in the dark.

If the solution is explainable and auditable, the risk of wrong decisions and further ramifications is lower. It is better for improvements in the business.